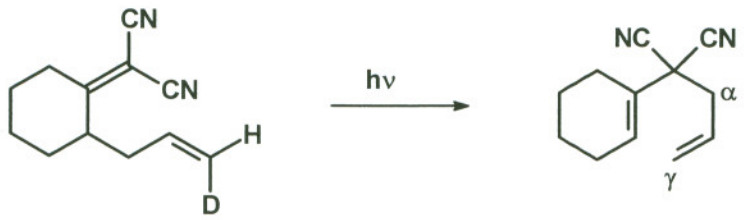
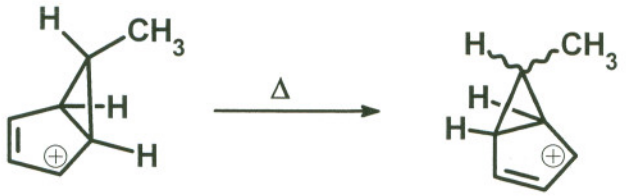
Organic Cume, November 9, 2007

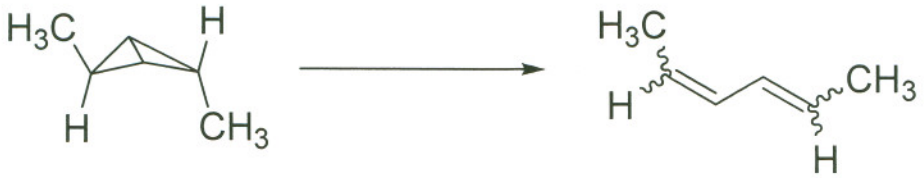
1. In the following photochemical conversion, designate the expected position of deuterium in the product (assume retention at the migrating group). Do you expect deuterium to be found at the α or γ position, or both? Why? (Explain using appropriate symmetry rule arguments)



2. In the following reaction, predict the configuration of the product shown. Why? (Explain using appropriate symmetry rule arguments).

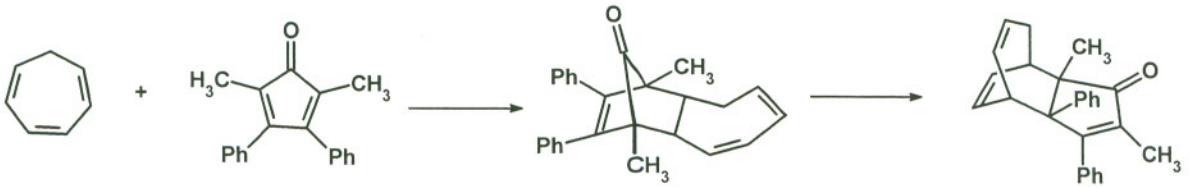


3. Predict, using orbital symmetry rules (show your work), the configuration of the product in the following thermal reaction.



Consider the below reaction scheme. When compounds **1** and **2** are reacted thermally, compound **3** is formed. Compound **3**, in turn, gives compound **4** in the following thermal reaction.

4. Predict the stereochemistry of compound **3**. Explain using appropriate arguments.
5. Give the mechanism for the formation of compound **4** from **3**. Give the name of this reaction.



1 **2** **3** **4**